

Diet, microbiota, and microbial metabolites in colon cancer risk in rural Africans and African Americans



Background: Epidemiologic studies have suggested that most cases of sporadic colon cancer can be attributed to diet. The recognition that colonic microbiota have a major influence on colonic health suggests that they might mediate colonic carcinogenesis. Objective: To examine the hypothesis that the influence of diet on colon cancer risk is mediated by the microbiota through their metabolites, we measured differences in colonic microbes and their metabolites in African Americans with a high risk and in rural native Africans with a low risk of colon cancer. Design: Fresh fecal samples were collected from 12 healthy African Americans aged 50–65 y and from 12 age- and sex-matched native Africans. Microbiomes were analyzed with 16S ribosomal RNA gene pyrosequencing together with quantitative polymerase chain reaction of the major fermentative, butyrate-producing, and bile acid–deconjugating bacteria. Fecal short-chain fatty acids were measured by gas chromatography and bile acids by liquid chromatography–mass spectrometry. Results: Microbial composition was fundamentally different, with a predominance of *Prevotella* in native Africans (enterotype 2) and of *Bacteroides* in African Americans (enterotype 1). Total bacteria and major butyrate-producing groups were significantly more abundant in fecal samples from native Africans. Microbial genes encoding for secondary bile acid production were more abundant in African Americans, whereas those encoding for methanogenesis and hydrogen sulfide production were higher in native Africans. Fecal secondary bile acid concentrations were higher in African Americans, whereas short-chain fatty acids were higher in native Africans. Conclusion: Our results support the hypothesis that colon cancer risk is influenced by the balance between microbial production of health-promoting metabolites such as butyrate and potentially carcinogenic metabolites such as secondary bile acids.